

Mediaeval: News Image Recommendation System

1. Problem Understanding

- ❖ Task: Recommend the most relevant image for a news article
- ❖ Inputs: article title (and optionally lead text)
- ❖ Output: one image per article
- ❖ Evaluation: Human judgment (relevance, non-misleading, editorial quality)
- ❖ Key Insight: This is a ranking/recommendation problem, not just generation

2. Dataset Preparation

- ❖ Obtain dataset from MediaEval (after agreement)
- ❖ Extract fields: article_id, title, optional lead text
- ❖ Prepare external image dataset (YFCC100M / similar)
- ❖ Store in structured format (CSV/JSON)

3. Preprocessing

- ❖ Remove null/empty titles
- ❖ Clean text (HTML decoding, remove noise)
- ❖ Normalize text (lowercase, strip spaces)
- ❖ Optional: sentence simplification

4. Baseline Retrieval Model

- ❖ Use CLIP model
- ❖ Encode all dataset images into embeddings
- ❖ Encode article titles into embeddings
- ❖ Compute similarity (cosine similarity)
- ❖ Retrieve top-K images per article

5. Keyword & Semantic Extraction

- ❖ Use KeyBERT or spaCy
- ❖ Extract entities and important terms
- ❖ Filter non-visual/abstract words
- ❖ Keep objects, places, people

6. Prompt Engineering Module

- ❖ Strategy 1: Basic keyword prompt
- ❖ Strategy 2: Structured semantic prompt
- ❖ Strategy 3: LLM-enhanced prompt generation
- ❖ Add style constraints (editorial, symbolic, minimal)

7. Image Generation Module

- ❖ Use Stable Diffusion (HuggingFace Diffusers)
- ❖ Generate multiple images per article (3–5)
- ❖ Control parameters: steps, guidance scale
- ❖ Use negative prompts to avoid noise

8. Hybrid Ranking System

- ❖ Combine retrieved and generated images
- ❖ Score all candidates using CLIP
- ❖ Optional heuristics: Penalize photorealism or Prefer symbolic/editorial styles
- ❖ Select best image

9. Post-processing

- ❖ Resize images to required size (460x260)
- ❖ Save in required naming format
- ❖ Ensure consistent formatting

10. Evaluation

- ❖ Automatic: CLIP score
- ❖ Human-aligned checks: Relevance and non-misleading content
- ❖ Compare: Retrieval vs Generation vs Hybrid

11. Experiments

- ❖ Run multiple configurations:
 1. Retrieval only
 2. Generation only
 3. Hybrid system
- ❖ Measure average scores
- ❖ Perform ablation study